

LAURINBURG-MAXTON, NC, USA

WMO: 723109

Lat: 34.792N

Lon: 79.366W

Elev: 220

StdP: 14.58

Time Zone: -5.00 (NAE)

Period: 99-19

WBAN: 93782

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	21.1	25.0	4.1	6.8	29.1	8.9	8.8	33.2	23.2	52.8	20.3	50.1	3.7	50	0.421

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.3	95.9	75.7	93.3	75.3	91.2	74.8	79.3	89.6	78.2	88.1	77.3	86.8	8.2	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
76.7	140.4	83.1	75.4	134.2	81.9	74.7	131.2	81.2	42.9	89.8	41.7	88.4	40.7	86.9	87.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
19.4	17.1	15.3	14.7	99.2	4.2	2.7	11.7	101.2	9.3	102.7	7.0	104.2	4.0	106.2		
			12.9	81.3	3.8	1.9	10.2	82.7	8.0	83.8	5.8	84.8	3.1	86.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	62.7	43.2	47.0	53.5	62.5	70.8	78.1	80.5	79.6	74.0	63.3	53.0	46.5
	DBStd	15.22	10.43	9.07	9.07	7.87	6.47	4.45	3.61	4.12	5.52	8.05	8.56	9.32
	HDD50	744	264	154	70	6	0	0	0	0	0	5	64	181
	HDD65	2799	676	509	369	138	25	0	0	0	7	129	370	576
	CDD50	5397	55	69	179	380	644	844	947	916	720	416	153	74
	CDD65	1976	1	4	13	63	204	394	482	451	277	75	9	3
	CDH74	18516	4	28	134	635	1957	3797	4793	4236	2228	629	64	11
	CDH80	7828	0	1	14	151	733	1729	2254	1936	854	153	3	0
Wind	WSAvg	6.5	6.5	7.1	7.8	7.9	7.1	6.3	5.9	5.2	6.0	5.9	6.0	6.0
Precipitation	PrecAvg	45.6	3.4	3.3	3.7	3.0	3.0	4.6	4.9	5.1	5.0	3.3	3.0	2.9
	PrecMax	55.9	6.9	6.9	6.1	6.0	7.4	9.6	8.6	11.0	12.8	8.9	7.9	5.2
	PrecMin	32.0	0.9	1.5	0.7	0.7	1.5	0.6	1.7	1.2	1.0	0.0	0.6	0.8
	PrecStd	5.8	1.9	1.4	1.4	1.4	1.3	2.4	2.0	2.2	3.0	2.2	1.9	1.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	73.5	78.5	82.0	88.1	95.0	97.8	98.8	99.1	94.1	89.1	80.0	75.4
		MCWB	63.8	64.8	64.4	68.4	72.2	74.6	77.0	77.4	73.1	73.8	67.2	66.2
	2%	DB	69.6	72.4	77.9	84.0	90.3	94.1	95.5	94.9	91.2	83.8	75.3	71.3
		MCWB	60.4	60.6	63.2	66.6	70.9	74.4	76.6	76.2	72.9	69.9	63.8	63.9
	5%	DB	64.6	67.8	73.8	81.2	87.6	91.5	93.0	92.0	88.3	80.8	72.3	67.0
		MCWB	57.9	57.7	59.9	65.1	70.2	73.9	76.4	76.2	73.1	67.8	62.6	61.8
	10%	DB	60.9	63.6	69.9	77.7	84.0	89.4	90.5	89.7	85.2	77.2	68.5	63.2
		MCWB	55.2	55.7	58.4	63.7	69.0	73.5	75.8	75.6	72.0	67.4	60.7	57.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	66.5	66.9	67.3	72.6	75.7	78.8	81.2	81.8	77.9	76.4	70.4	68.8
		MCDB	70.7	74.4	75.5	80.9	84.9	89.7	92.4	90.6	86.9	81.6	74.8	72.8
	2%	WB	62.9	64.1	65.4	70.0	73.7	77.5	79.5	79.4	76.5	74.0	67.5	65.7
		MCDB	67.4	69.7	73.9	79.1	83.8	88.4	90.4	88.9	84.8	80.5	71.7	69.1
	5%	WB	59.8	60.7	63.4	68.0	72.4	76.4	78.3	78.2	75.4	71.9	65.2	62.7
		MCDB	63.5	65.9	69.5	76.8	82.4	86.4	88.6	87.6	83.1	77.0	69.6	66.2
	10%	WB	55.7	56.8	61.4	66.0	71.1	75.4	77.4	77.2	74.3	69.4	62.5	58.9
		MCDB	59.7	62.3	68.1	74.3	80.5	84.4	87.1	86.2	81.1	74.7	67.3	62.9
Mean Daily Temperature Range	5% DB	MDBR	20.6	22.0	23.1	23.6	22.5	20.2	19.3	19.0	19.5	22.5	23.7	21.0
		MCDBR	24.5	26.6	27.4	27.1	26.0	23.5	22.6	22.3	23.3	25.1	26.1	23.4
		MCWBR	16.8	16.8	14.1	12.3	9.5	7.6	7.2	7.3	8.4	12.1	15.7	16.7
	5% WB	MCDBR	21.1	24.4	22.6	21.8	20.8	19.4	19.3	19.0	18.1	17.9	19.9	20.8
		MCWBR	16.7	17.9	14.2	11.5	8.5	7.4	7.4	7.3	7.3	9.9	14.3	16.8
Clear-Sky Solar Irradiance	taub	0.319	0.330	0.367	0.403	0.463	0.492	0.539	0.513	0.429	0.393	0.339	0.314	
	taud	2.492	2.454	2.383	2.277	2.160	2.118	2.007	2.070	2.301	2.363	2.475	2.541	
	Ebn at Noon	282	291	287	280	264	255	242	246	264	264	271	277	
	Edh at Noon	27	31	36	42	48	50	56	51	39	34	27	24	
All-Sky Solar Radiation	RadAvg	852	1058	1402	1757	1924	2022	1923	1740	1481	1218	934	742	
	RadStd	49	116	107	121	154	137	120	83	125	151	76	62	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	+2.14	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (68 neighbors)	+0.88	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+302	+216

Nomenclature: See separate page